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Objective Measures of Laryngeal Function After Reinnervation of the Anterior and Posterior Recurrent Laryngeal Nerve Branches

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Previous research indicates that separate reinnervation of the anterior and posterior branches of the recurrent laryngeal nerve (RLN) can provide purposeful motion of the larynx, even after transplantation. This canine study was undertaken to better determine the results of RLN reinnervation after nerve transection distal to its bifurcation. This approximates ideal conditions for transplantation, because potential rejection and nerve branch mismatch are eliminated. Eight months after nerve repair, video, electromyographic, mechanical, and histologic data were collected on four canines. Results show return of appropriate motion without synkinesis, including purposeful abduction on endotracheal tube occlusion. Abductory function was weaker on the reinnervated side, but adduction was equal or stronger on the reinnervated vocal cord. These results indicate that this method of RLN reinnervation produces consistent, strong physiologic motion in the denervated larynx.
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INTRODUCTION

In the past reinnervation of the larynx has been used to provide an injured or paralyzed recurrent laryngeal nerve (RLN) with a new source of efferent nerve fibers. Currently, because of the interest in reinnervation of the transplanted larynx, new approaches are being devised. We have found that the ability to fully and consistently rehabilitate the larynx after transplantation depends on the separate reinnervation of the laryngeal abductors and adductors. In this sense, the anatomy of the RLN is helpful, because in most larynges, the posterior branch of the RLN divides from the remainder of the nerve trunk and supplies the posterior cricoarytenoid, the only dilator of the larynx. The remaining RLN forms the anterior division, supplying each of the laryngeal adductors, with the exception of the cricothyroid. By reinnervating the RLN distal to its division, the uninterrupted pathway from the central nervous system to the RLN division is maintained, allowing both voluntary and reflex control of the larynx.

Previous attempts to reinnervate the RLN used a variety of methods, including direct neurorrhaphy[1] or grafts such as the vagus[2] and phrenic[3] nerves. The ansa cervicalis has also been used, either by anastomosis to the RLN stump or with a nerve-muscle pedicle.[4] Results have often been unpredictable, with a high incidence of failure of reinnervation and synkinesis.[5–6] Such methods may improve laryngeal function, but do not allow the coordinated purposeful motion that is present in the normal larynx. Because the majority of nerve fibers in the RLN are destined for laryngeal adductors, the small numbers of abductor nerve fibers are likely to be reinnervated with fibers that close the larynx. This may cause synkinesis wherein the larynx closes with respiratory effort, producing airway obstruction.

A previous study from this laboratory looked at transection and reanastomosis of the anterior (adductor) branch of the RLN, with use of a nerve-muscle pedicle in the denervation and reanastomosis of the abductor fibers.[7] The results using these methods were promising, with resultant appropriate vocal cord motion without synkinesis, but involved a single animal. The same method of reinnervation was used in earlier efforts at laryngeal transplantation. Three canines developed purposeful opening and closure of the transplanted larynx.[8]

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This study was performed in accordance with the PHS Policy on Humane Care and Use of Laboratory Animals, the NIH *Guide for the Care and Use of Laboratory Animals,* and the Animal Welfare Act (7 U.S.C. et seq.); the animal use protocol was approved by the Institutional Animal Care and Use Committee of the University of California, Los Angeles.

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TABLE I.
Tensionometry: Percentage Change of Left Versus Right.

	Experimental Dogs		Sham/Control Dogs*	
	Mean	SD	Mean	SD
Posterior cricoarytenoid, tube occlusion*	29.81	18.85	17.85	4.62
Posterior cricoarytenoid, posterior branch stimulation	55.52	15.37	38.68	16.05
Thyroarytenoid, RLN stimulation	24.69	27.99	136.43	126.47
Thyroarytenoid, anterior branch stimulation	2.58	19.03	—	—

*Includes only sham dog. There are no data available for sham/control dogs for thyroarytenoid, anterior branch stimulation.
RLN = recurrent laryngeal nerve.

The current study was undertaken to expand the scope of that study, and to look at the outcome with direct anastomosis of the posterior branch, without the use of a muscle pedicle. This is the first experiment to document laryngeal muscle strength after reinnervation with the "tensionometer" force gauge developed in our laboratory.[9]

MATERIALS AND METHODS

Experimental Model

Six male mongrel dogs (range, 25 to 35 kg) were studied. One dog served as control, with no surgery performed. One dog underwent a sham operation in which an incision was made, the anterior and posterior branches of the recurrent laryngeal nerve were identified on the left side and the wound was then closed without nerve division. The remaining four dogs underwent transection with immediate reanastomosis of the anterior and posterior branches of the RLN on the left side.

Methods

Each canine was anesthetized with intravenous pentobarbital and placed supine on an operating table. Direct laryngoscopy was performed to confirm normal laryngeal anatomy. The animal was intubated with a 7.0 endotracheal tube under direct visualization and kept under general anesthesia with inhalation of halothane in the range of 1% to 1.5%, with 100% O_2 at 4 L/min. The animal's neck was then shaved, prepped, and draped in a sterile fashion.

A midline incision was made from the level of the upper border of the thyroid cartilage to just below the cricoid cartilage. The strap muscles were separated along the midline and retracted laterally. The anterior and posterior branches of the RLN were isolated on the left side. The superior laryngeal nerve was identified and preserved. The inferior border of the cricopharyngeus was cut to allow access to the distal portion of the RLN at the point of bifurcation into anterior and posterior branches. These branches were transected and then reanastomosed with 10-0 nylon using microsurgical technique (Fig. 1).

The animals were recovered and skin sutures were removed at approximately 2 weeks. After at least 8 months (range, 8 to 11 mo) the animals were returned to the operating room for a second procedure at which time electromyographic data and results of vocal cord motion and force of motion were collected.

Following the same protocol the dogs were placed under general anesthesia. A distal tracheostomy was placed for ventila-

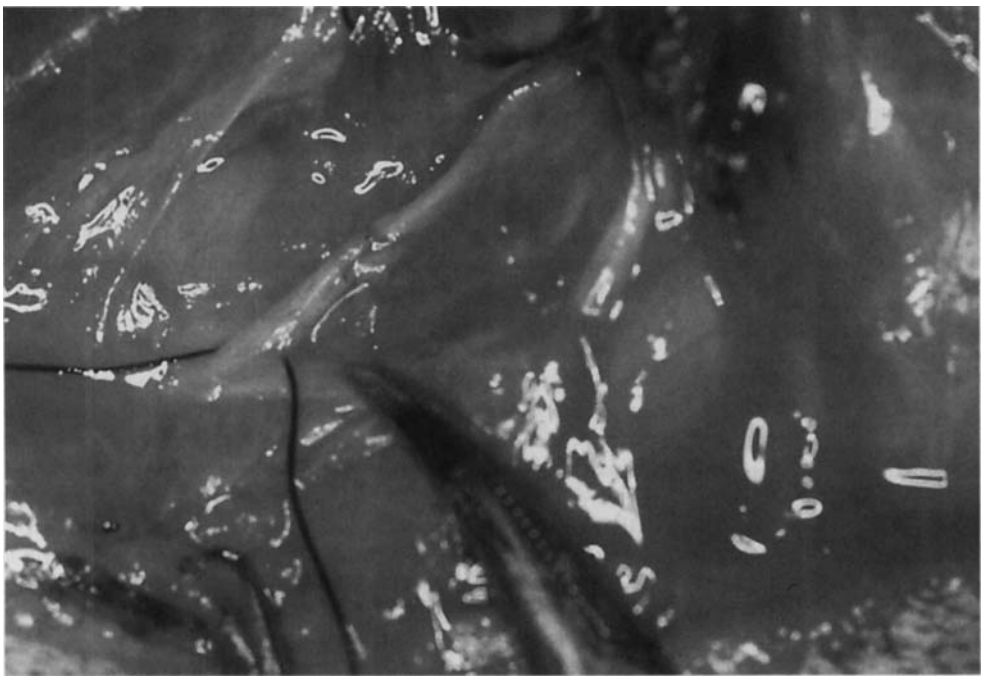


Fig. 1. The bifurcation of the recurrent laryngeal nerve into anterior and posterior branches. The suture identifies the anterior branch.

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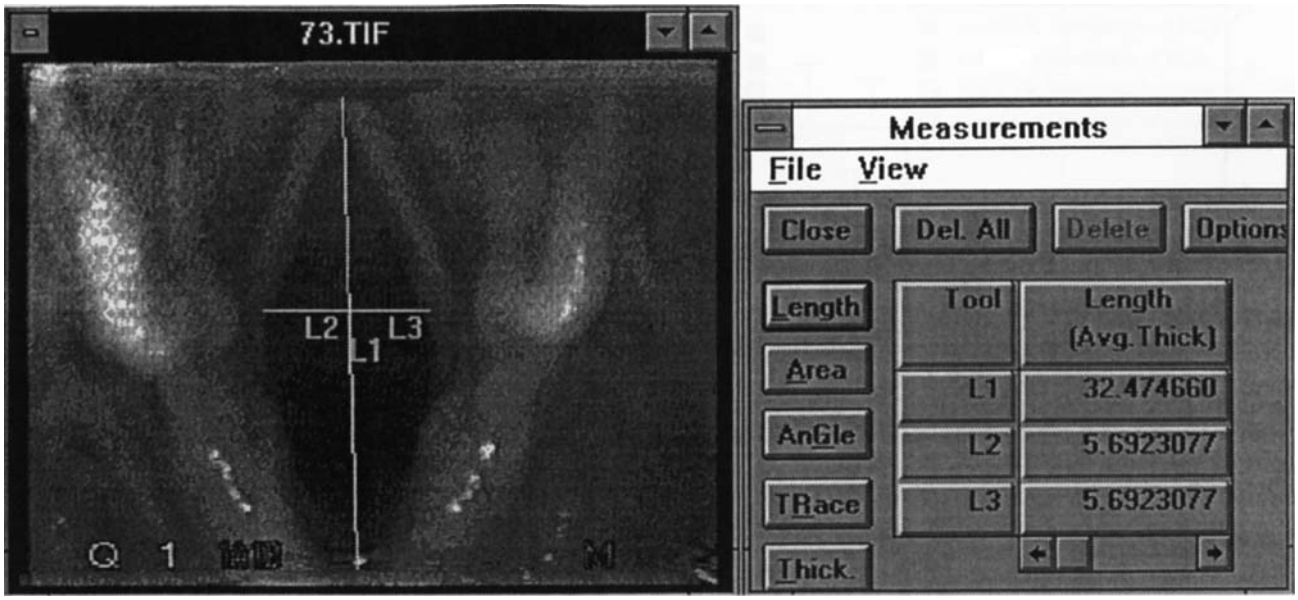


Fig. 2. Typical images and the measurements used for evaluation of vocal cord excursion.

tion; a mouth gag was used to allow direct visualization of the larynx. A button was placed through the epiglottis to suspend the epiglottis out of the field of view. Videolaryngoscopy and tensionometry were performed with obstructive respiratory dyspnea. Electromyography was performed under conditions of obstructive respiratory dyspnea and pharyngeal stimulation, and signals from the posterior cricoarytenoid and thyroarytenoid muscles, respectively, were recorded.

The RLN were dissected bilaterally and rubber electrodes (monopolar, flexible, conductive neoprene with silicone, coated with insulating silicone KE45 W) were applied. Videolaryngoscopy and tensionometry data were collected with nerve stimulation. Finally, the animal was humanely euthanized according to Animal Care and Use protocol, and tissue was harvested for histologic evaluation.

Equipment

For videolaryngoscopy, a zero-degree Storz endoscope was placed through the mouth and above the vocal cords. A video camera was used to record images on a Sony U-matic videocassette recorder, model VO-9850 (Sony, Japan). Analysis was performed using ImagePro for Windows 3.1 software (Media Cybernetics, Silver Spring, MD) on a Zeos 486 PC, using a FlashPoint ISA (Version 2.0, Integral Technologies, Indianapolis, IN) color frame grabber. Measurements were made of vocal cord excursion under the given conditions, comparing the relative excursion of each side against the midline. Measurements with nerve stimulation were compared from separate trials (Fig. 2).

Abductory force measurements were made by a "tensionometer," as described by Berke,[9] placing a straight rod with a plate with a surface area of 3.06 mm^2 just lateral to the midportion of

TABLE II.
Tensionometry: Absolute Force for Abduction and Adduction.

	Experimental Dogs		Sham/Control Dogs*	
	Mean	SD	Mean	SD
Posterior cricoarytenoid, tube occlusion				
Right	2.20	1.04	3.54	0.53
Left	1.63	1.03	2.86	0.31
Posterior cricoarytenoid, posterior branch stimulation				
Right	3.58	0.74	5.57	0.96
Left	1.64	0.70	3.28	0.34
Thyroarytenoid, RLN stimulation				
Right	3.87	1.24	2.38	0.89
Left	4.83	1.50	4.62	0.88
Thyroarytenoid, anterior branch stimulation				
Right	4.96	0.98	—	—
Left	4.98	0.79	—	—

*Includes only sham dog. There are no data for sham/control dogs for thyroarytenoid, anterior branch stimulation.

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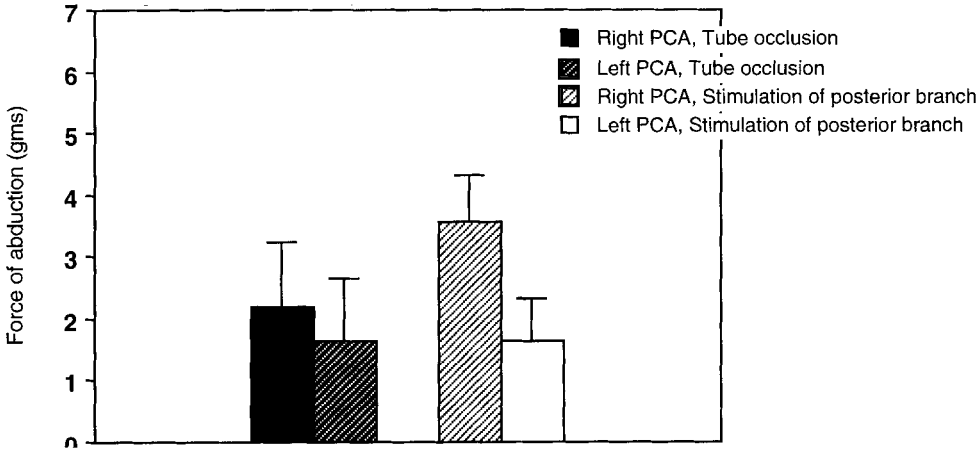


Fig. 3. Experimental dogs. Force of abduction, right versus left, under varying conditions.

the arytenoid. The tensionometer was adapted with a digital force gauge (DF-0.5R, Shimpo, Japan). Adductory measures were made using the same rod, placing the plate medial to the true vocal cord, just anterior to the anterior border of the arytenoid. Comparative measurements were made for abductory and adductory force under each set of conditions. Five sequential measures of abduction or adduction were made for each trial. Zero and peak readings were recorded, and the zero settings were subtracted.

A constant current nerve stimulator was used (Model S2LH, WR Medical Electronics Co., St. Paul, MM) for RLN stimulation. Frequency and duration of stimuli were 80 Hz and 1.5-ms duration, respectively. Stimulus levels for abduction measures were done by finding the threshold level for maximal abduction. Adductory measures were done at the threshold for complete glottal closure. Stimulus levels ranged from 4 to 190 mA.

Electromyography was performed using the Nicolet Viking IV system (Nicolet, Madison, WI). Concentric needle electrodes (26 gauge, 1.5 inches long, recording area 0.068 mm^2; Nicolet, Madison, WI) were used, placing the needles into the respective muscles via a transoral approach.

For histology, tissue was preserved in 10% formalin. The anterior and posterior branches of the recurrent laryngeal nerve were then embedded in plastic for electron microscopy using standard techniques.[10] Sections were cut at points proximal and distal to the anastomotic site and stained with toluidine blue. Morphometry was performed to assess the total number of fibers as well as the number of large and small fibers.

All animal protocols were approved by the UCLA Animal Care and Use Committee and were performed in compliance with the local, state, and federal regulations for the humane use of animals in research.

Statistical Analysis

Statistical analysis was performed using a two-way ANOVA, dog by side, with repeated measures on side under each set of conditions. Experimental dogs were compared within and between subjects, and sham and control dogs were assessed separately. Tensionometry data were assessed, comparing abductory and adductory force on sides under conditions of nerve stimulation and obstructive respiratory dyspnea. Glottal images obtained from nerve stimulation were analyzed to compare vocal cord excursions on each side. Data were analyzed using SYSTAT software (SPSS, Inc., Chicago, IL).

RESULTS

Tensionometry

Results of tensionometry for abduction are shown in Figures 3 and 4. Abduction was weaker on the operated side, although the force on this side represented approximately 70% of the normal side for obstructive respiratory

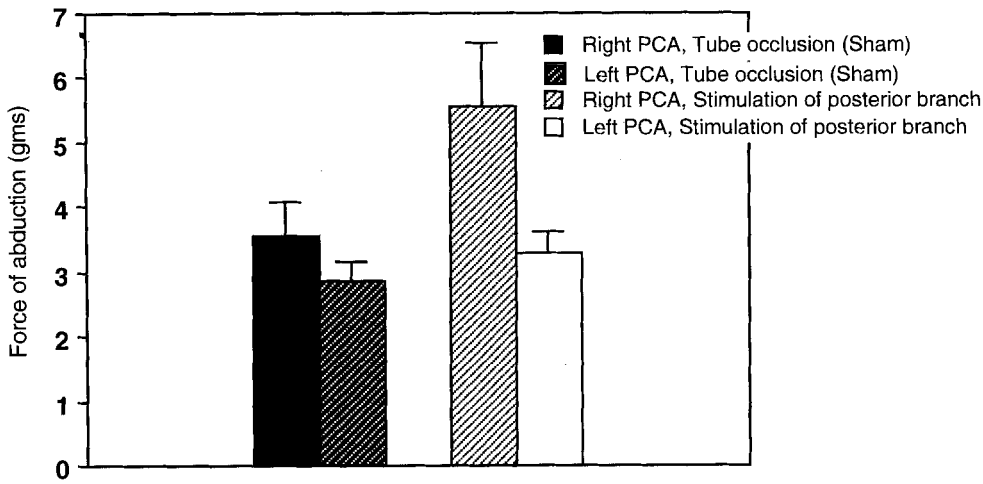


Fig. 4. Control and sham dogs. Force of abduction, right versus left, under varying conditions.

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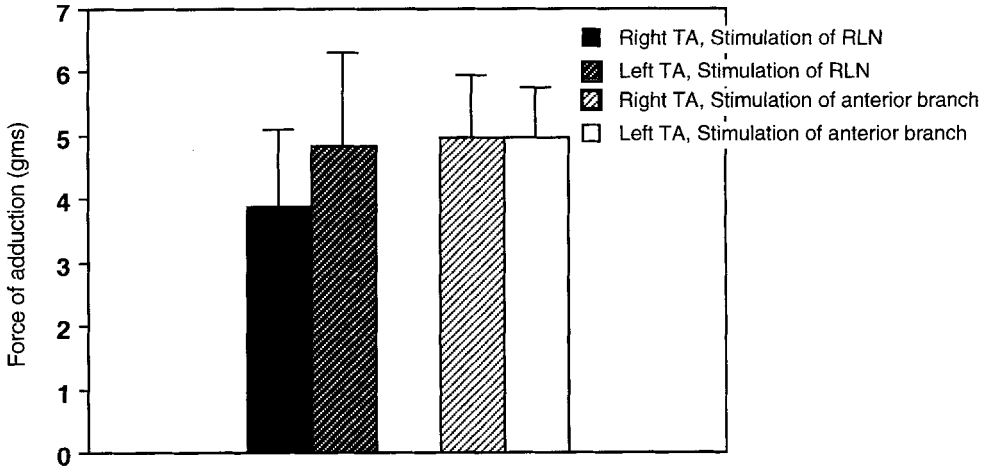


Fig. 5. Experimental dogs. Force of adduction, right versus left, with stimulation of main trunk of the recurrent laryngeal nerve versus adductor branch.

motion. The left side was weaker than the right with nerve stimulation as well, with the left side producing only 44% of the force compared with the right, approximately the same absolute force under both sets of conditions. Results were statistically significant at a level $P < .05$ for a difference between sides with obstructive respiratory dyspnea, $F(1, 16) = 66.99$. Differences between sides for nerve stimulation were significant at a level $P < .05$, $F(1, 16) = 938.99$. Control and sham dogs showed similar results with respect to left and right. The left side produced force equal to approximately 80% of the right with tube occlusion, and 59% of the right with nerve stimulation. Significance between sides for tube occlusion was $P < .05$, $F(1, 4) = 38.64$, with these results including only the control dog. For posterior branch stimulation, there was a difference between sides for the control and sham dogs, with $P < .05$, $F(1, 8) = 537.86$ (Table I).

Adductory force measurements show that stimulation of the anterior branch of the recurrent laryngeal nerve resulted in nearly equal force in comparisons of the operated and nonoperated sides (Table II). There was no statistically significant difference in sides under these conditions. Stimulation of the main trunk of the RLN resulted in an unequal adductory force, with the nonoperated side approximately 25% weaker than the reinnervated side. Under this set of conditions, there was a statistically significant difference between the operated and nonoperated sides, $P < .05$, $F(3, 16) = 153.6$. The sham and control dogs show similar results for stimulation of the RLN, with the left being stronger than the right, $P < .05$, $F(1, 8) = 484.79$ (Figs. 5 and 6).

Vocal Cord Excursion

Analysis of the videolaryngoscopy images under varying conditions shows similar excursions of right versus left. For the experimental dogs, excursions of right versus left with tube occlusion are 53.1% and 46.9%, respectively, with a standard deviation of 4.33%. With stimulation of the abductor branch of the RLN, the excursions increase to 64.2% (SD, 10.83%) and 62.89% (SD, 10.68%), respectively. The

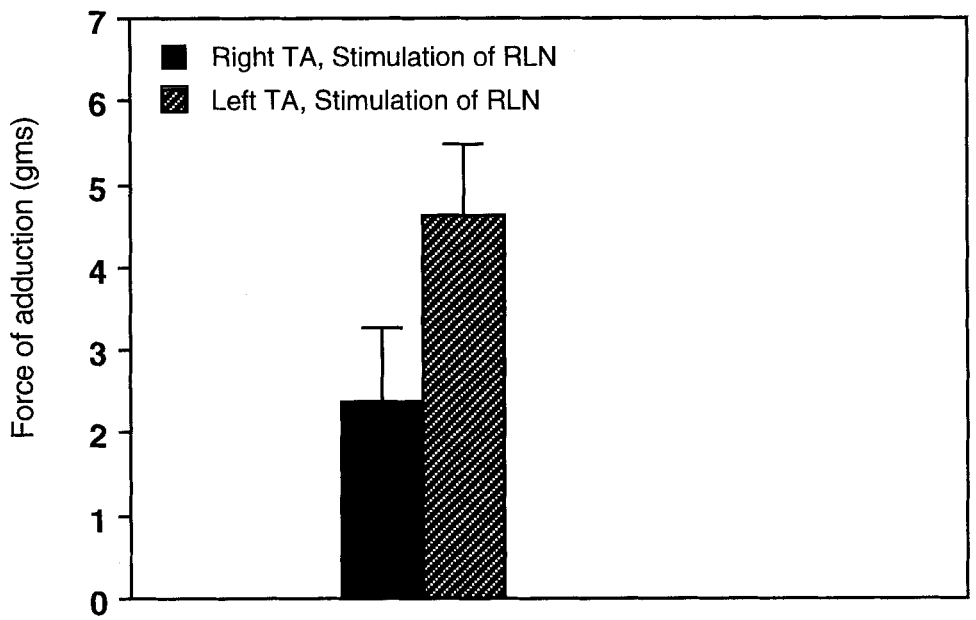


Fig. 6. Control and sham dogs. Force of adduction, right versus left, with stimulation of the recurrent laryngeal nerve.

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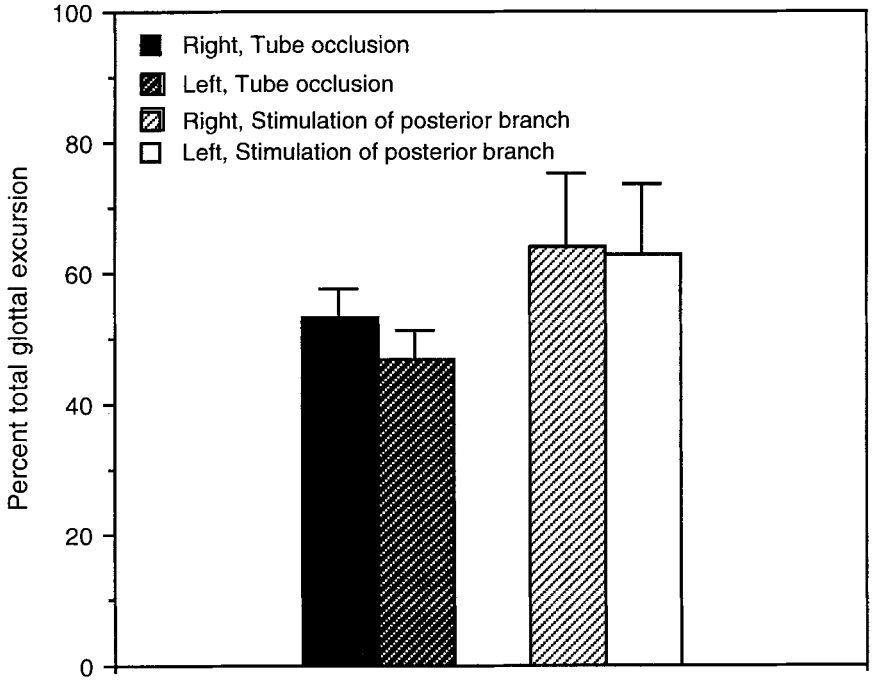


Fig. 7. Experimental dogs. Comparison of excursion with abduction, right versus left, under varying conditions.

sham dog shows excursions of 53% and 47%, right versus left (SD, 1.58%), which increase to 64.75% (SD, 6.90%) and 60.4% (SD, 3.85%), respectively, with stimulation of the posterior branch of the RLN (Table III). These results show no statistically significant effect of side, with $P > .50$, $F(1, 16) = .51$ (Figs. 7 and 8).

Electromyography

Electromyography results demonstrate firing of polyphasic and normal-appearing motor units for the posterior cricoarytenoid with respiration, and the thyroarytenoid with pharyngeal stimulation. In addition, nerve conduction trials demonstrate signals in both the thyroarytenoid and posterior cricoarytenoid muscles with stimulation of the main trunk of the RLN. These results indicate appropriate reinnervation of the respective muscles (Fig. 9).

Histology

Histologic evaluation, with comparison of fiber counts from sections proximal and distal to the anasto-

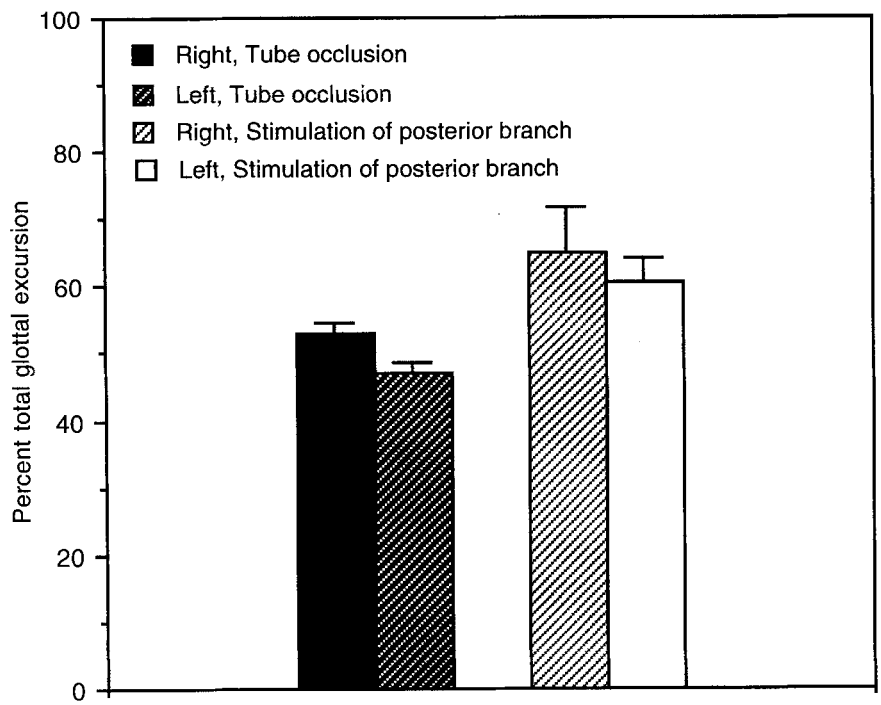


Fig. 8. Control and sham dogs. Comparison of excursion with abduction, right versus left, under varying conditions.

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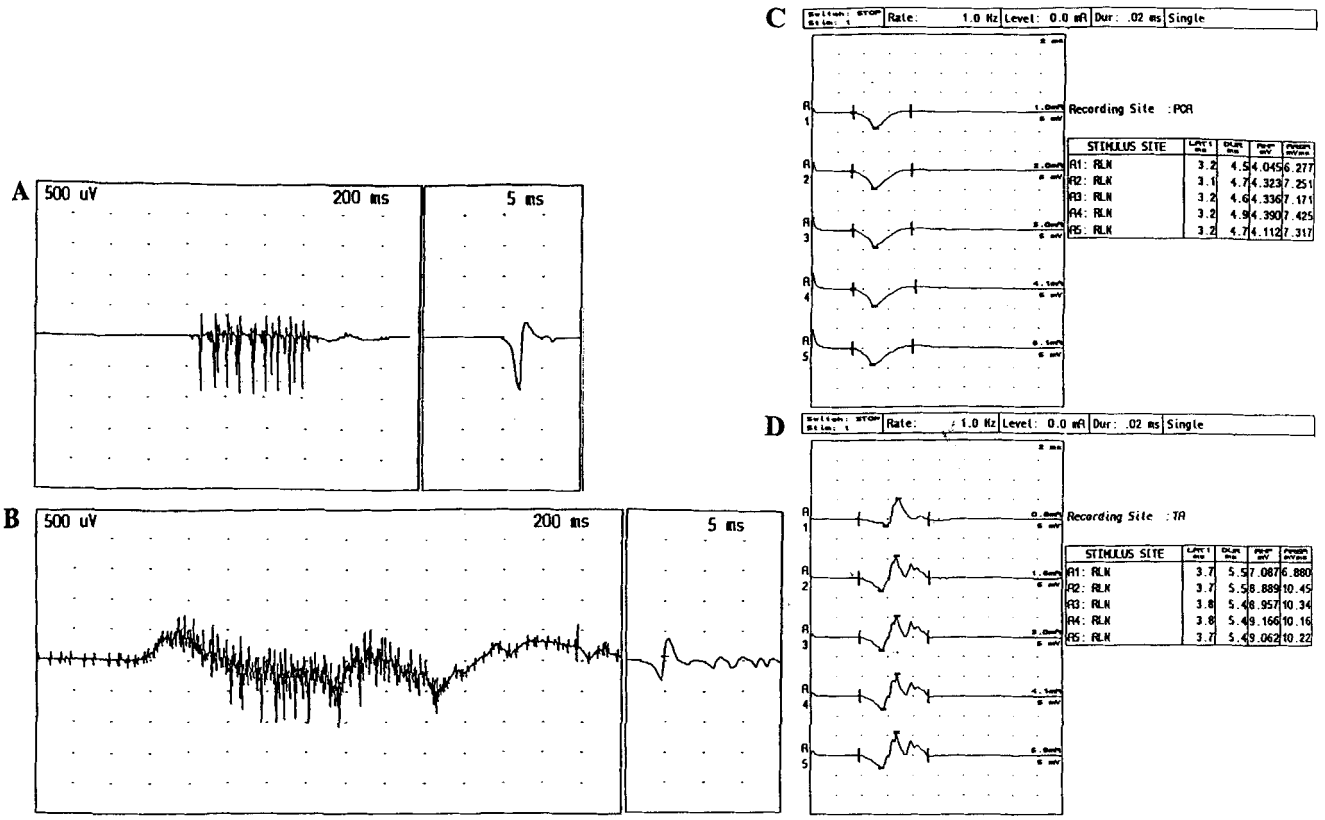


Fig. 9. **A.** Electromyographic signal from the posterior cricoarytenoid with obstructive respiratory dyspnea. Signal corresponds to inspiration. Expanded view shows a normal-appearing motor unit. **B.** Electromyographic signal from the thyroarytenoid with pharyngeal stimulation. Increased firing of motor units corresponds to adduction. Expanded view shows typical motor unit. **C.** Nerve conduction study of posterior cricoarytenoid, with stimulation of the main trunk of the recurrent laryngeal nerve, demonstrating continuity. **D.** Nerve conduction study of thyroarytenoid, with stimulation of the main trunk of the recurrent laryngeal nerve, also showing a response.

motic sites, demonstrates maintenance or recovery of a large number of fibers distal to the anastomosis (Fig. 10). The anterior branch of subject 73, distal to the anastomosis site, had 89% of the number of fibers compared with the proximal segment (Table IV). The anterior branch of animal 74 had an equal number of fibers in both proximal and distal portions. The posterior branch of this subject had 96% of the number of fibers in the distal segment compared with proximal. Histologic sections were not usable from all subjects secondary to artifact from staining and sectioning. Control and sham dogs were not studied.

TABLE III.
Image Analysis: Comparison of Changes in Glottal Area and Vocal Cord Excursions.

	Experimental Dogs		Sham/Control Dogs*	
	Mean (%)	SD (%)	Mean (%)	SD (%)
Change in area with abduction				
Tube occlusion	53.25	24.59	84.41	2.68
Posterior branch stimulation	105.36	44.71	212.41	33.80
Vocal cord excursion				
Tube occlusion				
Right	53.10	4.33	53.00	1.58
Left	46.90	4.33	47.00	1.58
Posterior branch stimulation				
Right	64.20	10.83	64.75	6.90
Left	62.89	10.68	60.40	3.85

*Data only from sham dog.

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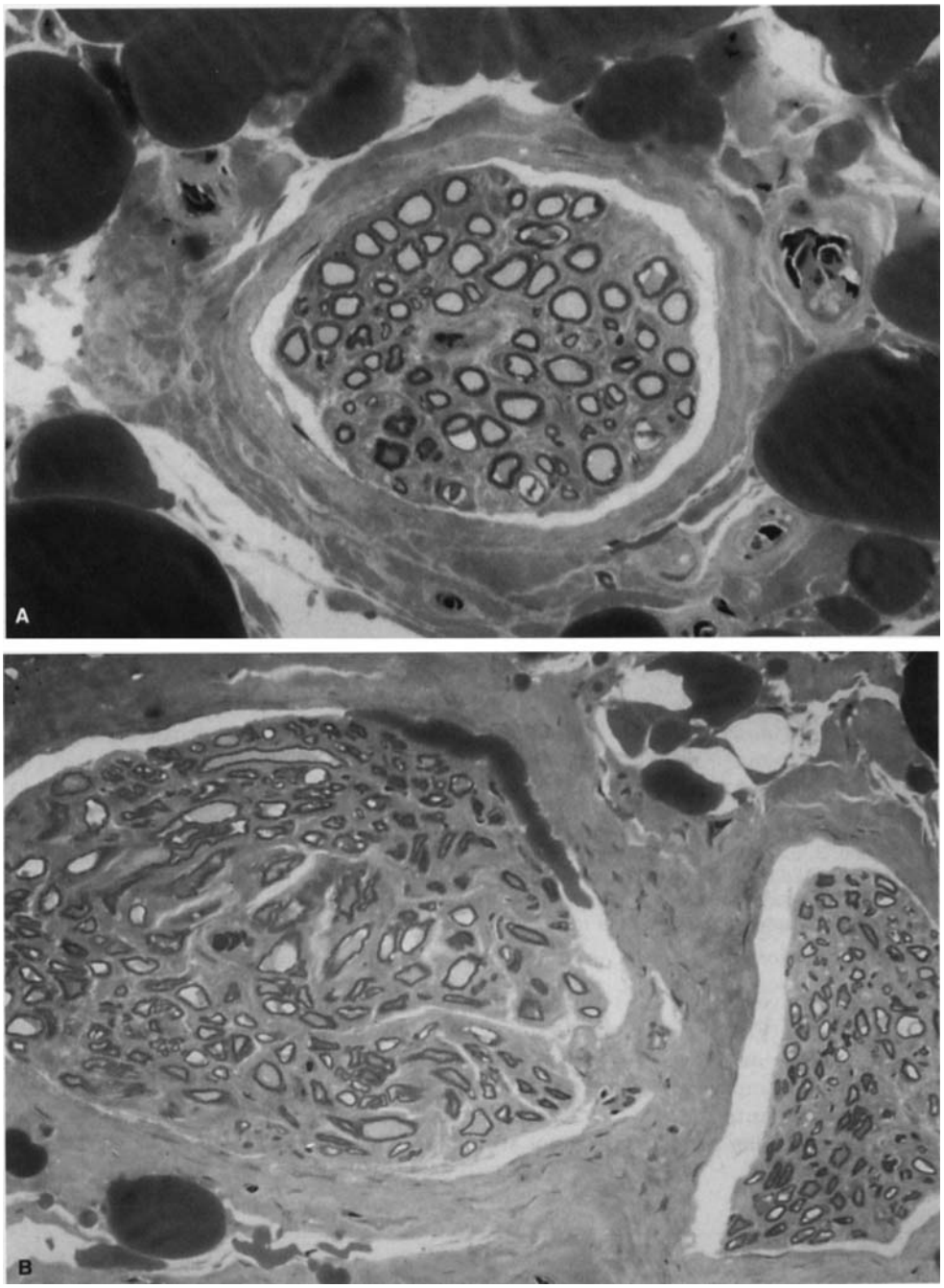


Fig. 10. Histologic sections from the anterior branch of the recurrent laryngeal nerve in subject 75. **A.** Proximal segment. **B.** Distal segment. (Toluidine blue, magnification × 40)

DISCUSSION

Physiologic motion indicates the ability of the larynx to open and close appropriately during laryngeal tasks. This includes dilatation of the larynx with respiratory efforts and closure during swallowing or phonation. In this study, each animal attained physiologic motion, with particularly strong adduction—the force produced was equal to the normal side. While the posterior cricoarytenoid did reinnervate, the strength of contraction was less than the contralateral, unoperated side. This is consistent with previous studies in our laboratory in which abduction of the reinnervated larynx was present but weak.

This appropriate motion of the larynx is not possible if the RLN is reinnervated with a mixed population of adductory and abductory fibers, as with the use of the RLN trunk. Previous results for direct neurorrhaphy of the RLN trunk have been mixed. For example, Gordon and McCabe[1] demonstrated return of adduction only after direct neurorrhaphy, attributing loss of abductory motion to a prevalence of adductory fibers. Doyle et al.[11] reported on a canine study involving immediate repair of the recurrent laryngeal nerve versus the use of a split vagus graft. Their results were quite optimistic, with "nearly 100 per cent success." They had consistent reinnervation, but vocal cord motion could be elicited only by electrical stimulation. No spontaneous physiologic motion was attempted. Tashiro[5] investigated direct neurorrhaphy and found inconsistent function 1 year later. He

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TABLE IV.
Histology.

Subject	Fiber Total	Large Fibers	Distal/Proximal
73			
Anterior proximal	320	55.33	
Anterior distal	285	62	89.1%
Posterior proximal	Lengthwise sections	Uncountable	
Posterior distal	Lengthwise sections	Uncountable	—
74			
Anterior proximal	202.2	34.33	
Anterior distal	204	40.3	100.9%
Posterior proximal	333	106	
Posterior distal	321	141	96.4%
75			
Anterior proximal	345	76	
Anterior distal	230	56	66.7%
Posterior proximal	Uncountable secondary to staining artifact		
Posterior distal	143	56	—
77			
Anterior proximal	241	68	
Anterior distal	One fascicle uncountable	Incomplete count	—
Posterior proximal	370	80	
Posterior distal	One fascicle uncountable	Incomplete count	—

concluded that these results were most likely due to misdirected reinnervation and significant enough to consider reinnervation a failure.

Synkinesis can be a significant problem in reinnervation attempts.[12] This results from misdirected fibers inappropriately reinnervating antagonistic muscles,[6] with subsequent paradoxical movement with respiration and airway compromise. Adductory fibers outnumber abductory fibers by a ratio of 4:1[13] such that adduction predominates on reinnervation. Therefore, some investigators have attempted selective reinnervation of the abductory muscles in order to maximize airway function.

Tucker described a nerve-muscle pedicle technique of posterior cricoarytenoid reinnervation using a branch of the ansa hypoglossi to a strap muscle, typically the omohyoid. Studies first in a canine model[14–15] and later in humans[16] showed good results, with return of abductory function in all patients. Results were adequate to allow decannulation of three of five patients, without significant loss of vocal function. This technique has not been widely adopted because of inconsistent results at other centers.

Murakami and Kirchner[17] demonstrated adequate vocal cord abduction after reinnervation by selective denervation of the adductor branches after recovery of function. Although they were able to obtain appropriate respiratory abduction, they also noted abduction on stimulation of the laryngeal afferents, allowing abduction triggered by swallowing. Despite this unwanted abduction, the animals in their study showed no signs of aspiration.

The ability to reinnervate the larynx with consistency is essential in the development of laryngeal transplantation. Although an immobile larynx may allow speech, few potential patients would accept the risk of immunosuppression for such a result when the alternative of tracheoesophageal speech is available. This study was undertaken in order to determine what outcome could be expected under presumably ideal conditions for laryngeal reinnervation, using our canine laryngeal transplantation model. These conditions consist of immediate reanastomosis to optimize timing and a perfectly size-matched RLN, transected distal to the bifurcation of the RLN into adductor and abductor branches. This minimizes the possibility of synkinesis.

Although the posterior nerve branch is small, a direct anastomosis is possible in most canine and human larynges. In laryngeal transplantation, this would allow a more complete laryngeal resection than a posterior cricoarytenoid nerve-muscle pedicle because it would not be necessary to preserve any intrinsic musculature from the recipient larynx.

Results from the present study show partial return of function on the operated side, allowing appropriate abduction and adduction. Abductor function recovered to nearly normal in terms of excursion, and reached approximately 71% of the force of abduction of the right side under conditions of obstructive respiratory dyspnea. Forty-five percent of abductory force, as compared with the unoperated side, was recovered under conditions of stimulation of the posterior branch of the RLN. Comparing the results obtained in the experimental animals with the

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sham and control dogs, these results may represent nearly total return of function. Adduction was even stronger on the operated side, possibly because a weaker posterior cricoarytenoid provided less inhibition of closure.

Unexpectedly, the sham and control animals showed a sidedness, such that the left side was weaker both with selective stimulation and after obstructive respiratory dyspnea. This is possibly owing to the small number of control animals. This sidedness has not been encountered in our previous canine research.

None of the animals demonstrated any evidence of synkinesis. Each animal showed appropriate abduction under conditions of obstructive respiratory dyspnea. Abductory and adductory motion was produced independently with nerve stimulation. Attempts at pharyngeal stimulation to elicit spontaneous adductory motion were inconsistent, requiring the animals to be under light anesthesia. Electromyography results showed appropriate activity for the posterior cricoarytenoid and thyroarytenoid under conditions of obstructive respiratory dyspnea and pharyngeal stimulation, respectively. Nerve conduction studies demonstrated appropriate conduction from the main trunk of the RLN to the posterior cricoarytenoid and thyroarytenoid muscles. The histologic data corroborate the physiologic findings. Nerve fiber regeneration studies show the number of distal fibers to be in the range of 66.7% to 100% of the proximal segment.

These results are quite promising, especially considering the force produced on the reinnervated side of the larynx, and suggest that with appropriate match of the nerve branches and no significant ischemic injury, rejection, or infection, laryngeal transplantation can produce a functional larynx. All dogs demonstrated good return of abductory and adductory function. In order to achieve return of function with appropriate adductory and abductory motion, RLN anastomosis should be done distal to the bifurcation into respective antagonistic branches.

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